

YIFAN XU

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Education

University of Illinois at Urbana-Champaign

Master Science in Information Management GPA 4.0/4.0

Aug. 2024 – Present

Champaign, Illinois

ShanghaiTech University

Bachelor of Engineering in Computer Science

Sep. 2020 – June 2024

Shanghai, China

Research Experience

Tracing LLM Reasoning Processes with Strategic Games

Graduate Research Intern

University of Illinois at Urbana-Champaign

Feb. 2025 – May. 2025

- Designed and developed the "Turn-based Attribute Game" for the AdvGameBench framework, a complex environment with elemental attributes and skill budgets to test the strategic planning and decision consistency of LLMs.
- Evaluated 12 state-of-the-art LLMs within the game, analyzing process-level metrics like Correction Success Rate (CSR) and Over-Budget Rate (OBR) to quantify their ability to plan, revise, and manage resources under constraints.

NanoNeRF: Nanoscale 360° reconstruction with NeRF under SEM

Undergraduate Research Intern

Oct. 2023 – Jan. 2024

ShanghaiTech University

- Invented new super resolution and 3D reconstruction techniques to enhance the precision and speed of controlling nanorobots, typically observed under a Scanning Electron Microscope (SEM), where imaging quality and speed are inversely related.
- Achieved rapid reconstruction of nanotubes captured under SEM by modifying the Neural Radiance Field's pinhole imaging model to parallel projection using **Pytorch**, aligning it with the physical imaging process of SEM, improving average PSNR from 22.3 to 28.13.

Review of Deep-learning-based Super-Resolution

Undergraduate Research Intern

Feb. 2023 – Sep. 2023

ShanghaiTech University

- Conducted extensive research on blind and non-blind super-resolution methods, from classification to application scenarios, and authored a comprehensive review paper.

Publications

- Xiang Fu, **Yifan Xu**, Hu Su*, Song Liu, "NanoNeRF: Robot-assisted Nanoscale 360° reconstruction with neural radiance field under scanning electron microscope", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* and Selected by **Oral Representation**, Abu Dhabi, Oct. 2024.
- Ying Li, **Yifan Xu**, Hu Su, and Song Liu, "A Review of Deep-learning-based Super-Resolution: from methods to applications", in *Pattern Recognition*, paper id: 10.1016/j.patcog.2024.110935.

Teaching Experience

CS100: Introduction to Programming

Teaching Assistant

Spring 2023

Shanghai, China

- Hosted weekly office hours to tutor students on fundamental C/C++ programming concepts and assist with debugging assignments.
- Graded programming assignments, quizzes, and exams, providing constructive feedback to reinforce students' understanding.

Open Source Experience

VizSeed - Domain Specific Language for VChart and VTable | *TypeScript*

June 2025 - Present

- Designed and implemented a domain-specific language (DSL) that abstracts complex multi-dimensional data visualization into **declarative syntax**, reducing chart configuration complexity by **80%** and enabling **seamless chart type switching** (bar, line, pie, table) without rewriting specifications.
- Architected a **data-driven visualization pipeline** that transforms raw datasets into standardized VizSeed format through **dimension-measure abstraction**, supporting **arbitrary dimensional data visualization** and **cross-platform chart rendering** for both VChart and VTable engines.

Technical Skills

Languages: Python, Java, HTML/CSS, TypeScript, JavaScript, SQL, Markdown, Shell, R, MATLAB

Developer Tools: Docker, AWS, VS Code, Linux/Unix, MySql, Microsoft SQL, MongoDB, PostgreSQL

Technologies/Frameworks: Flask, Node.js, React, Numpy, Pytorch, Mongooose, Express, Hadoop, Spark, Git